

DNS Service Incident & Network Traffic Analysis

1. Project Overview

Conducted a simulated cybersecurity incident investigation involving DNS service disruption and network connectivity failures affecting customer access to public-facing website.

The project focused on:

- network traffic analysis,
- Incident documentation,
- Protocol investigation,
- Root cause assesment,
- Operational impact evaluation,
- Escalation and remediation planning.

Network logs were analyzed using tcpdump to identify communication failures related to DNS services and UDP traffic.

2. Incident scenario

Customers reported inability to access a company website and received the error : „destination port unreachable”. The investigation identified repeated failures involving DNS communication over UDP port 53, resulting in service disruption and degraded website availability.

3. Investigation activities

- reviewed tcpdump network traffic logs
- Analyzed DNS, UDP and ICMP traffic behavior,
- Identified failed communication attempts to DNS port 53
- Interpreted ICMP error responses indicating unreachable services
- Evaluated potential causes including: DNS server outage, firewall misconfiguration, denial-of-service conditions and service unavailability.
- Documented finding in formal incident report format.
- Escalated findings for further remediation investigation.

4. Key findings

The analysis identified:

- repeated UDP communication failures,
- ICMP „port53 unreachable” responses,
- DNS service availability issues,
- Potential operational disruption affecting customer-facing services.

The incident demonstrated how DNS failures can directly impact:

- service continuity,
- Customer experience,
- Business operations,
- Organizational availability objectives.

5. Governance & Risk perspective

The project emphasized:

- incident response documentation,
- Operational risk awareness,
- Service availability monitoring,
- Escalation procedures,
- Communication between security and engineering teams,
- Business continuity considerations.

6. Recommendations

Recommended next steps included:

- verifying DNS server availability,
- Reviewing firewall rules affecting port 53,
- Monitoring for denial-of-service indicators,
- Improving service monitoring capabilities,
- Strengthening incident response escalation procedures.

7. Skills Demonstrated

- incident analysis,
- Network traffic analysis,
- Dns investigation,
- Risk assessment,
- Incident documentation,
- Root cause analysis,
- Operational risk awareness,
- Tcpdump,
- ICMP/UDP Analysis
- Escalation procedures,
- Service availability analysis,
- Cybersecurity reporting